Project Proposal

Aiden Cathcart, Jiaqi Lei, Krishaant Pathmanathan, and Zane Williams

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Research Question

Does air quality have a statistically significant effect on income inequality within Canadian Metropolitan Areas?

Proposal Summary

This research project seeks to understand the relationship between air quality and income inequality within Canadian Census Metropolitan Areas (CMAs). Specifically, we aim to determine whether higher Air Quality Health Index (AQHI) scores are correlated with higher estimated Gini Coefficients within a CMA. By establishing this relationship, we seek to contribute to broader understandings of environmental justice and socioeconomic disparities in

Canadian urban spaces. This project is within the scope of Track 3: Applied Urban and Social Economics Research.

Methodology

Data Collection and Variables

All data used in this project will be collected from three sources:

1. Environment and Climate Change Canada's observation data from the Air Quality Health Index (AQHI) program;
2. Statistics Canada's 2021 Census boundary files;
3. and Environics Analytics' 2024 Demographic Estimates & Projections obtained through SimplyAnalytics and derived from Statistics Canada's 2021 data.

All data will be organized by CMA. As hourly AQHI measurements are only available from December 2024 to February 2025, we will only use numerical data from 2024. Spatial data will be sourced from 2021: CMA bounds have not changed since Statistics Canada's 2021 census.

Spatial data will be used to attribute AQHI measurement stations to their corresponding CMA. Not all AQHI measurement stations are located within a CMA, and not all CMAs contain an AQHI measurement station—these incomplete measurements will be excluded from our data. Notably, Quebec does not allow for the release AQHI measurement observations. In the event that a CMA contains multiple AQHI measurement stations, their daily averages will be combined.

From these data sources, we will extract the following dependent, independent, and control variables:

Dependent Variable: Estimated Gini Coefficient

The CMA demographic data does not include a measurement of income inequality. We will estimate the Gini Coefficient for each CMA population using the following 2024 demographic data:

1. I_0: Median Household Income
2. I_1: Percent of Households which have a Household Income of \$100,000 or higher
3. I_2: Percent of Households with a Household Income of \$100,000 or higher which have a Household Income of \$200,000 or higher

4. I_3: Percent of Households with a Household Income of $200,000 or higher which have a Household Income of $300,000 or higher

The Gini Coefficient is a measure of income inequality $G \in [0,1]$, with $G = 0$ representing perfect income equality and $G = 1$ representing maximum income inequality. We will estimate the Gini Coefficient as follows:

Let us first standardize our data. Let N represent the number of CMAs. For each CMA $k|k \in [1, N]$, we will expand our income vector I_k to a matrix IP_k which contains median-adjusted income and income percentile.

$$IP_k = \begin{bmatrix} 0 & 0.5 & \frac{I_1}{100} & \frac{I_1 I_2}{10000} & \frac{I_1 I_2 I_3}{1000000} \\ 0 & I_0 & \frac{100000}{I_0} & \frac{200000}{I_0} & \frac{300000}{I_0} \end{bmatrix}$$

The first row of IP_k represents the percentile of a given median-adjusted income value with terms x_n. The second row of IP_k represents the median-adjusted income value with terms y_n.

We now need to approximate the Lorenz curve for k. This will be completed using the Lagrange Interpolation Polynomial, which is

$$P(x) = \sum_{n=0}^{4} y_n L_n(x)$$

where $L_n(x)$ are the Lagrange basis polynomials of the form

$$L_n(x) = \prod_{j=0, j\neq n}^{4} \frac{x - x_j}{x_n - x_j}$$

We also have a median-adjusted line of income equality of the form $E(x) = \frac{y_5}{x_5} x$.

Our Gini coefficient can now be estimated by:

$$G(IP_k) = \frac{\int_0^{x_5} (E(x) - P(x))\, dx}{\int_0^{x_5} E(x)\, dx}$$

We will numerically approximate these integrals using Simpson's Method.

Independent Variable of Interest: Air Quality Health Index (AQHI)

The AQHI is a composite index which measures various air pollution levels (PM2.5, PM10, NO , CO, O , SO). Higher AQHI values indicate worse air quality. An AQHI value higher than 3 represents a risk to public health. While the AQHI is typically presented to the public as an integer value, hourly measurements are reported with a precision of two decimal places. We will interpret a CMA's AQHI data in one of two ways:

1. Average daily AQHI average;
2. or Percentage of daily AQHI averages with a value greater than 3.

Additional Regressors and Control Variables

Our control variables are sourced from Environics Analytics' 2024 Demographic Estimates & Projections. The following demographic and economic variables have been chosen to minimize Omitted Variable Bias. Our objective is to control for any correlation between our independent variable of interest and our error term to uphold Conditional Mean Independence. Adding sufficient control variables will isolate the effect of air quality on income.

Demographic Factors:

Variable & Purpose	Key Data & Findings	References
% Male & Female Population by Age • Account for gender-based income disparities	- Gender ratio in manufacturing areas: Higher proportion of males - National gender ratio: Stable (47.3%-52.6%), average ~50%	Varechova et al., 2016
% Population by Visible Minority Status • Assess racial wealth disparities	- Minority distribution: <22.4% in 78% of regions; >50% in some - Citizenship: 85.6% Canadian citizens - Higher % of visible minorities near industrial areas	American Lung Association, 2023; WHO, 2022
% Married or Cohabiting • Control for effect of household size	- Married/cohabiting families: >50% in most regions - Single-parent families often located near high-pollution areas due to affordability	Pais et al., 2014

Variable & Purpose	Key Data & Findings	References
% Knowledge of Official Languages • Evaluate language barriers	- English: Bimodal distribution (main peak 56%, secondary 81%) - French: <15.1% outside Quebec - Non-official languages (e.g., Chinese, Hindi): <13%	———
% First-Generation Immigrants • Understand wealth accumulation and income gaps	- Majority age group: 39-45 years - Slight aging trend in population distribution	———

Economic & Employment Factors:

Variable & Purpose	Key Data & Findings	References
% Household Income >$60K (After Tax) • Control regional wealth levels	- Median income: 76K–95K (right-skewed) - High-emission industries in wealthy regions (e.g., Toronto: 20% of Canada's GDP)	PMC, 2023; Stockholm Environment Institute, 2023
% Labour Force Employed • Account for job market conditions	- Employment rate: 51.8%–60.3% - Structural unemployment in some regions	———
% Population (15+ Years) Without Education Credentials • Assess education's impact on income	- 13.8%–20.1% of labor force lack higher education - Limited career choices for this group. - Manufacturing/mining workers often do not have high rates of university-level education	———

Regression Model

Our regression will have the form:

$$\text{Gini Coefficient} = \beta_0 + \beta_1(\text{AQHI}) + \beta_2(\text{Gender Income Disparity}) + \beta_3(\text{Visible Minority Status}) +$$
$$\beta_4(\text{Marital Status}) + \beta_5(\text{Language Knowledge}) + \beta_6(\text{Generation Status}) +$$
$$\beta_7(\text{High Income Population}) + \beta_8(\text{Employment Rate}) + \beta_9(\text{No Education Credentials}) + \epsilon$$

Descriptive Statistics

Figure 1: Canadian Census Metropolitan Areas | Percentage of Days with AQHI > 3 (2024)

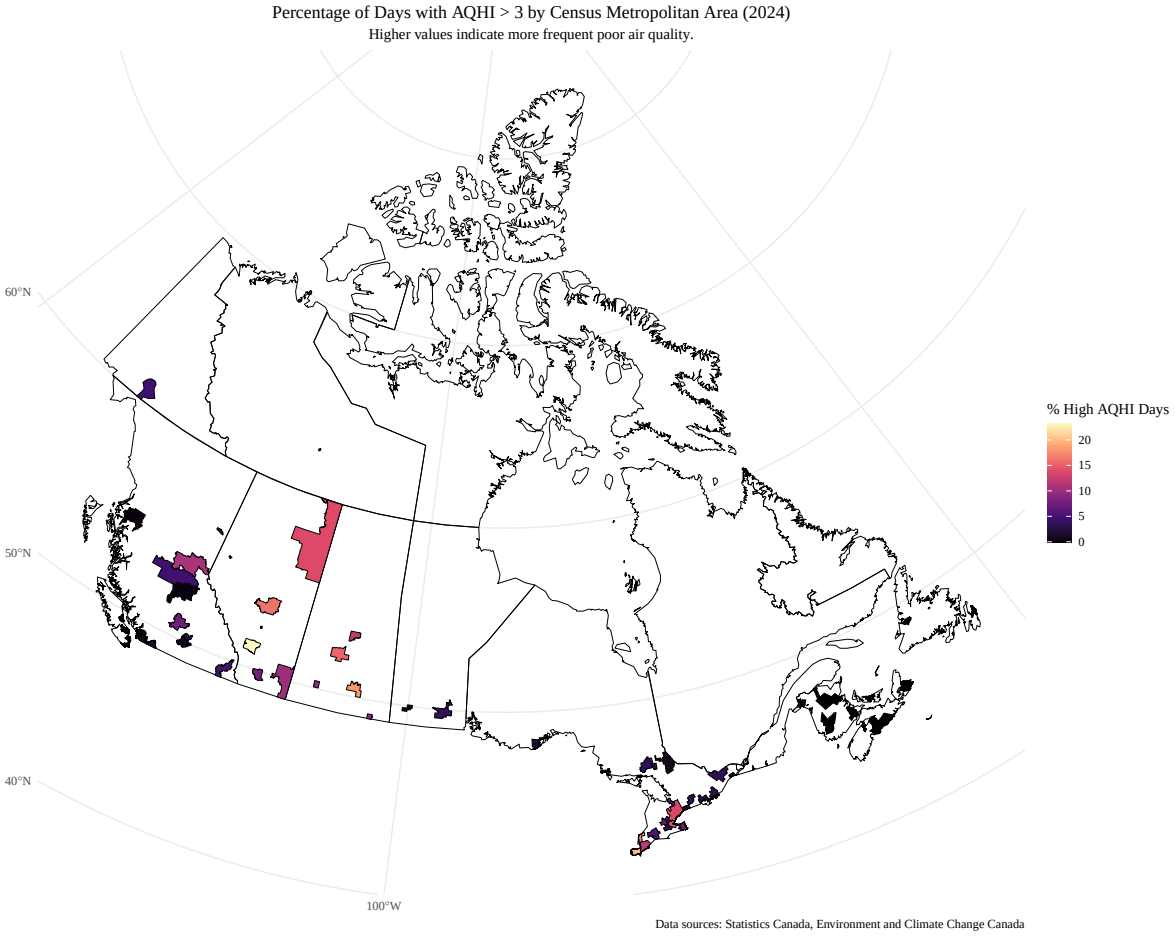


Figure 2: Canadian Census Metropolitan Areas | Average AQHI (2024)

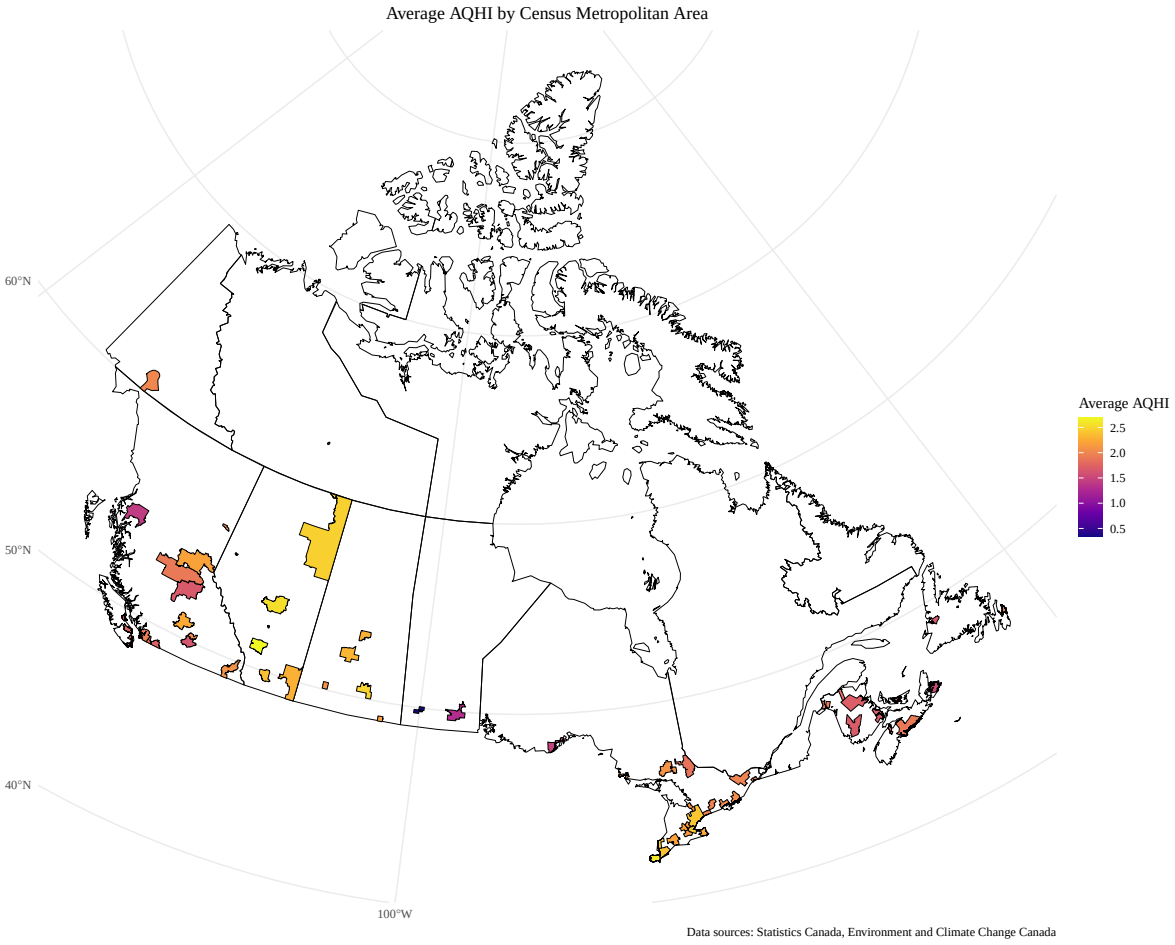
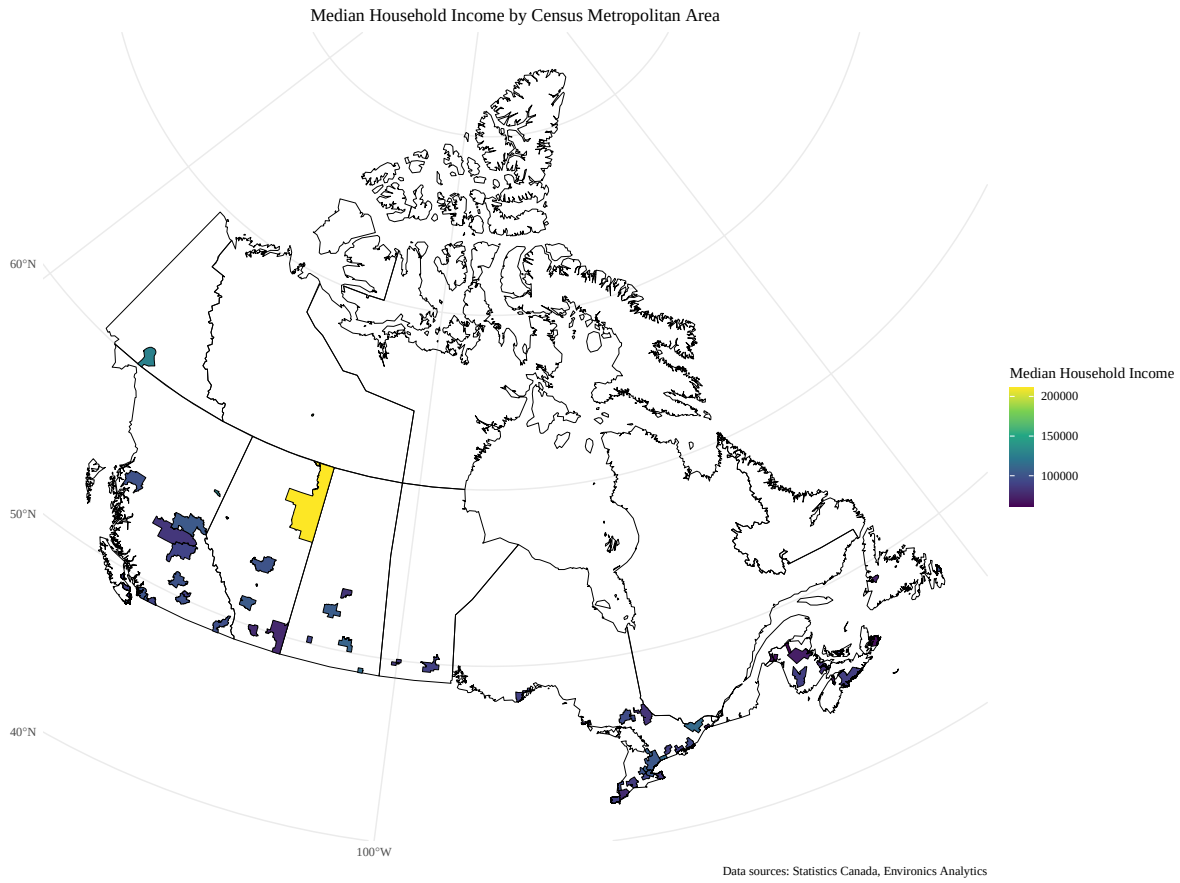


Figure 3: Canadian Census Metropolitan Areas | Median Household Income (2024)



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